

LATENT OCEAN

Data that organizes itself.

Diren Kumaratilleke · direnavk@outlook.com

www.latent-ocean.com

THE PROBLEM

Today's data systems chase what's common and miss what matters. The most valuable patterns hide in the rare events, the unusual signals, the connections nobody is looking for.

Data is multiplying faster than anyone can understand it.
The next generation of systems has to organize itself.

INGESTION

Read any kind of data which can be structured or messy, big or small.

FOLLOW THE FLOW

Don't just see what's there and be capable of tracking how data patterns move.

REORGANIZE ITSELF

Build its own structure as it learns. No engineer required.

WHAT IT IS

A new way to read data;
one that lets every kind of information speak the
same language.

Lets every kind of data line up against every other.

LESS NOISE

REUSABLE PARTS

ONE SHARED
LANGUAGE

Two things, attached to every row.

48
BITS

A FINGERPRINT

A barcode for every row of data. Two rows that are similar get similar barcodes. Two rows that are different get very different ones. This is how the system knows what belongs together, across millions of rows.

4
NUMBERS

AN UNUSUALNESS SCORE

Four numbers per row. How predictable it is, how rare its position is, how anomalous it looks, and how all of that combines into one signal. The score tells you whether a row is ordinary, or worth a closer look.

BEFORE & AFTER

What rows looked like. What rows look like now.

BEFORE · JUST ROWS

ID	DATE	AMOUNT
A-1024	Mar 12	\$340,000
A-1025	Mar 14	\$415,000
A-1026	Mar 18	\$398,000
A-1027	Mar 21	\$372,000
A-1028	Mar 25	\$450,000

Same as every other row. Nothing tells you which one matters.

AFTER · STRUCTURALLY AWARE

ID	FINGERPRINT	SCORE
A-1024	11010110...	0.21
A-1025	11010101...	0.18
A-1026	11011010...	0.24
A-1027	00101110...	0.91
A-1028	11010111...	0.22

A-1027 surfaces immediately. Its fingerprint is unlike its neighbors and its score is high.

Five thousand filers in. *One stood out.*

TOP-1 OUT OF 4,999 PUBLIC FILERS

American Electric Power

Asset Retirement Obligation

INDEPENDENTLY VERIFIED · THREE SOURCES

DELOITTE

On the Radar · Jul 2025

Asset retirement obligations require “significant management estimates” and rank among the highest-scrutiny disclosures.

EY

2025 SEC Comment Letter Trends

AROs remain a recurring topic in SEC staff reviews of utility 10-Ks.

AEP 10-K

Annual Report · FY 2023

Cook nuclear decommissioning ARO of \$2.4B, plus material coal-ash retirement liabilities.

WHAT MAKES IT DIFFERENT

Most data systems are designed by hand. Latent Ocean grows on its own with no setup needed.

TODAY'S DATA

- Designed once
- Architecture frozen
- Domain-specific
- Brittle to new data

LATENT OCEAN

- Discovers its own structure
- Reorganizes itself
- Works across domains
- Grows with new data

WHERE IT APPLIES

One system. Many worlds.



FINANCE

Reading the SEC archive as one connected story and surfacing the companies that don't fit.



MEDICINE

Finding the connections across millions of medical papers that no practitioner has time to read.



RESEARCH

Connecting centuries of scientific thinking. Finding the threads no one person could see in a lifetime of reading.

AND TOOLS BUILT ON TOP · FOR EVERY OTHER FIELD

Real findings. Real domains.

1

Found rare, high-value patterns in messy real-world data. Patterns that line up with what experts already know, and reveal what they don't.

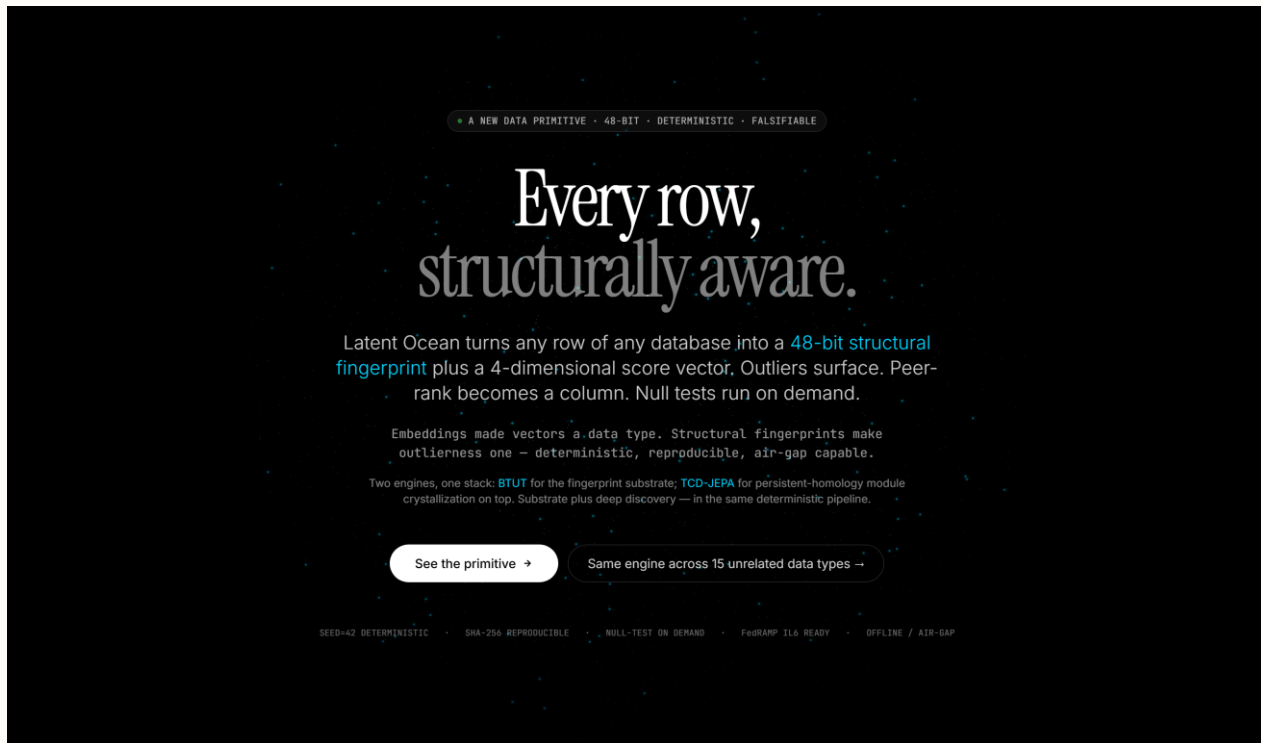
2

Reads live data and surfaces what matters most. Adjustable by region and timeframe.

3

Applied a single system to multiple entirely different fields, without retraining.

Every row, structurally aware.



The home page. What visitors see first.

Structural fingerprints. A first-class column.

THE PRIMITIVE · A DATA TYPE, NOT A PRODUCT

Structural fingerprints. A first-class column.

Every row that passes through Latent Ocean acquires two deterministic columns: a **48-bit structural fingerprint** and a **4-dimensional score vector**. The fingerprint locates the row in a compressed representation of its universe. The score vector reports how structurally unusual that location is. Both are reproducible at bit level under a fixed seed.

What it is

THE FINGERPRINT

48 bits. Deterministic rotation-ensemble hash of the row's structured attributes. Two rows whose signatures are close in Hamming distance are structurally similar. Two rows whose signatures are far apart are structurally divergent.

```
lo_fp: bit(48) = '1110110011011001101010101000110111010110100  
11101'
```

THE SCORE VECTOR

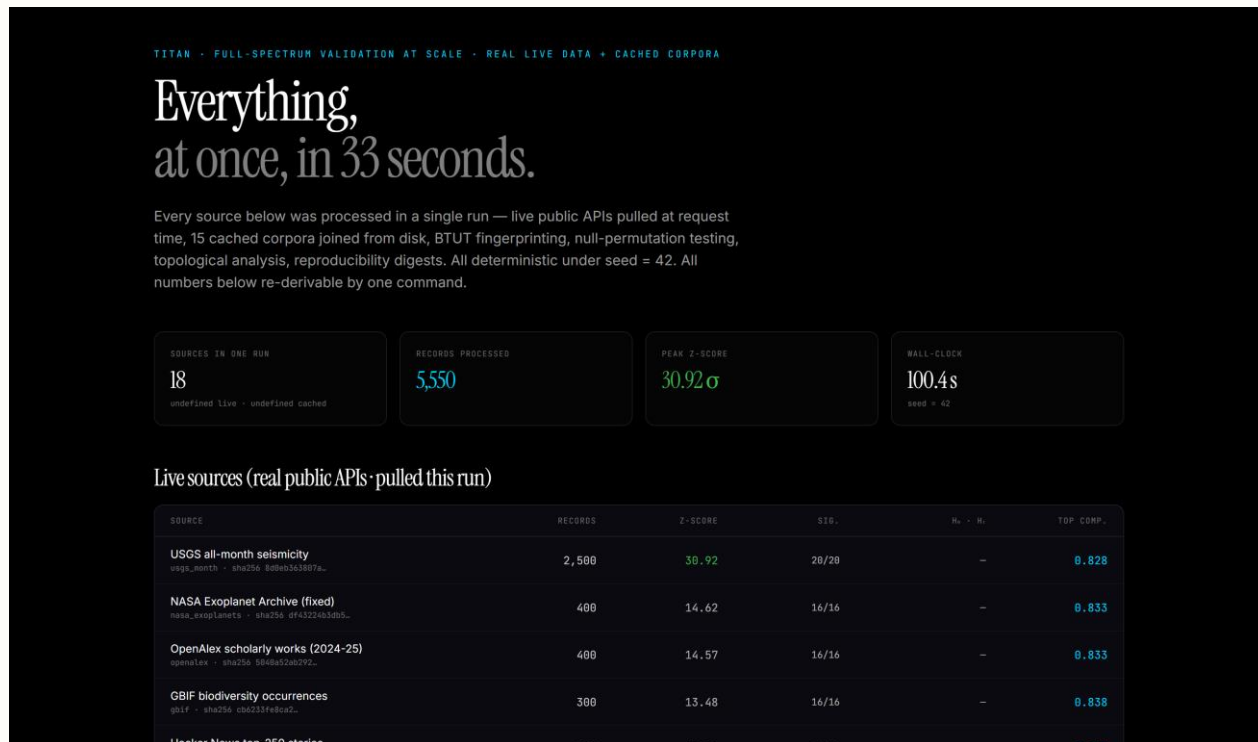
Four dimensions. Each is a normalized [0, 1] measurement against the rolling universe. The composite is a fixed weighted combination: 0.40-reconstruction + 0.35-diversity + 0.25-anomaly.

```
lo_score: {  
  composite: 0.859,  
  anomaly: 1.000,  
  reconstruction: 0.894,  
  diversity: 0.815  
}
```

What it replaces

The product, explained. One primitive, fifteen data shapes.

Everything, at once, in 33 seconds.



Eighteen public APIs, processed live, in a single deterministic run.

Real-time. Really real.

LIVE PIPELINES · REAL-TIME EXTERNAL FEEDS · UNIVERSAL FINGERPRINT PRIMITIVE

Refresh now

Real-time. Really real.

Two streaming sources — USGS global seismicity and CoinGecko top-100 markets — fetched live, hashed through the same 48-bit rotation-ensemble fingerprint, scored against a per-stream rolling history, and ranked by composite score. One engine, two radically different domains, both updating every 20 seconds.

USGS global seismicity

earthquake.usgs.gov ~ all_hour feed

Fetched
02:06:50

 Upstream
11ms

 Feed size
6

 History
20

LIVE · TICK 1

#	EVENT	MAG	RM	SCORE
#1	17 km ESE of Ninilchik, Alaska 10:01:25 · 11100110101...	1.7	53	0.888
#2	36 km ESE of Pāhala, Hawaii 10:30:40 · 11001001010...	1.9	42	0.793
#3	10 km W of Houston, Alaska 10:15:43 · 110001011101...	2.1	20	0.788
#4	6 km SSE of South Lake Tahoe, CA 09:58:46 · 010200011000...	2.2	13	0.689
#5	2 km WSW of The Geysers, CA 09:52:39 · 00011010101...	0.7	6	0.671
#6	6 km ESE of Carpinteria, CA 09:54:47 · 000010010010...	1.5	1	0.658

CoinGecko top-100 markets

api.coingecko.com ~ USD, cap-ranked

Fetched
02:06:50

 Upstream
62ms

 Feed size
100

 History
300

LIVE · TICK 1

#	ASSET	PRICE	24H	SCORE
#1	Toncoin TON mcap \$1.38 · 001010100101...	\$1.33	+0.9%	0.841
#2	Beldex BDx mcap \$420M · 01101001100...	\$0.080101	-0.1%	0.837
#3	Bitcoin Cash BCH mcap \$9.10 · 001010100111...	\$454.82	-0.9%	0.836
#4	POL (ex-MATIC) POL mcap \$99M · 110100011110...	\$0.093054	-3.0%	0.836
#5	Flare FLR mcap \$470M · 111000111111...	\$0.007848	-0.9%	0.831
#6	MemeCore M mcap \$5.48 · 010101010111...	\$4.33	-1.1%	0.831

Two streaming sources, scored every twenty seconds.

Trade secrets, with timestamps no one can fake.



- The deep technical methods are intended to be protected as trade secrets.
- Every invention is cryptographically time-stamped on a public blockchain.
- Anyone can verify the creation timestamp without ever seeing the invention itself.